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The circuit described here is of a dual audio-signal tracer. An audio-signal tracer is a simple device used to troubleshoot for audio signals in radio and other electronic circuitry.

Semiconductors:

- IC1 - LM358 dual op-amp
- IC2 - TDA2822M stereo op-amp
- D1-D5 - 1N4148 small-signal diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R9, R10 - 10-kilo-ohm
- R2 - 8.2-kilo-ohm
- R3, R4 - 1-kilo-ohm
- R5, R7, R13, R14 - 2-kilo-ohm
- R6, R8 - 18-kilo-ohm
- R11, R12 - 4.7-ohm
- R15 - 680-ohm
- VR1, VR2 - 1-mega-ohm potmeter

Capacitors:

- C2, C6, C7 - 100 μ F, 16V electrolytic
- C3, C9 - 0.15 μ F ceramic disk
- C4, C5 - 10 μ F, 16V electrolytic
- C8, C9 - 0.1 μ F ceramic disk
- C10, C11, C14 - 470 μ F, 16V electrolytic
- C12, C13 - 0.33 μ F ceramic disk

Miscellaneous:

- CON1-CON3 - 2-pin connector
- 6V battery
- 8-ohm, 0.5W speakers

LS1, LS2

The simple external diode detector is not only sensitive to amplitude modulation but even circuits that are normally used for other modulation schemes (such as FM radios) can be tested in some cases.

Fig. 1 shows the circuit diagram of the dual audio-signal tracer. It is built around LM358 (IC1) and TDA2822M (IC2). LM358 has two independent, high-gain, internal-frequency-compensated operational amplifiers.

TDA2822M is a low-power stereo amplifier. Its outputs can be directly coupled with the speakers through decoupling capacitors. The tracer is sensitive enough to capture signals from most microphones.

The relatively-high input resistance of 1-mega-ohm (VR1 and VR2) makes it possible to capture audible signals from many tone-control circuits and filters. Input signals are applied to connectors CON1 and CON2 and amplified by the two amplifiers in IC1.

Both amplifiers (channels) are completely independent. Resistor R3 and diodes D2 and D3 protect the input of the first amplifier. Similarly, resistor R4 and diodes D4 and D5 protect the input of the second amplifier. Resistors R1 and R2, diode D1 and capacitor C1 provide the necessary bias DC voltage for the operational amplifiers in IC1.

Potentiometers VR1 and VR2 adjust the volume for the first and second

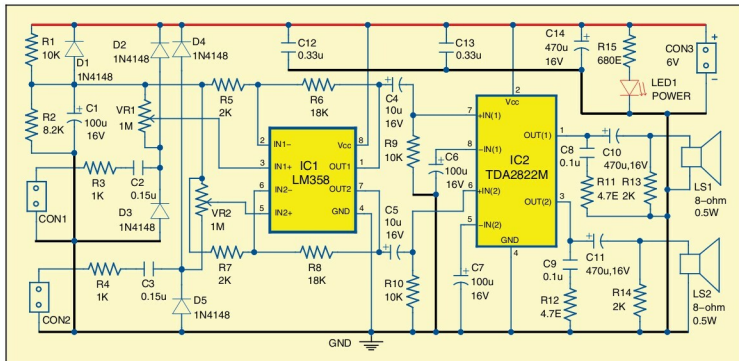


Fig. 1: Circuit diagram of the dual audio-signal tracer